f) The new law being implemented in December last year, where the back passengers have to belt up, _____ a large fall in the fatality rates during car accidents.

g) Because of the bad publicity, the company's share price _____ when the stock market opens again on Monday.

h) *Static Ventures* _____ (definitely) the job as its engineers have the experience and qualification to undertake the project.

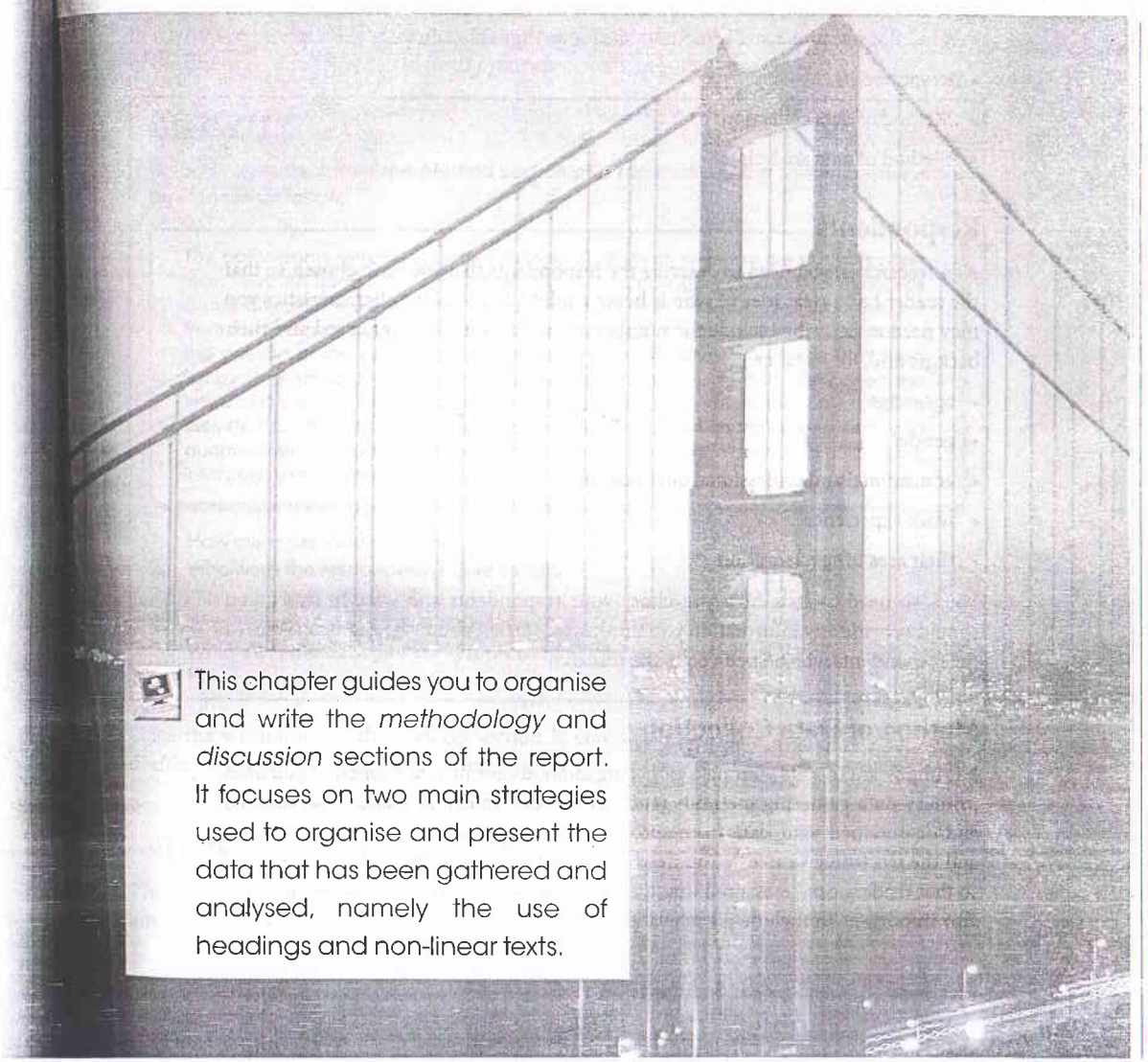
(i) The research _____ whether there is a need to improve the present training programme.

(j) The main objective of the investigation _____ to find out the reasons for the poor performance of the marketing department.

Source.
Sarguna et. al (2010). Report Writing for Business & Professional Purposes. KL: McGraw Hill p. 109-132

CHAPTER 8

Drafting the Report (II): Methodology & Discussion



This chapter guides you to organise and write the *methodology* and *discussion* sections of the report. It focuses on two main strategies used to organise and present the data that has been gathered and analysed, namely the use of headings and non-linear texts.

The body of the report constitutes two main parts: the methodology section which describes the methods used to gather and analyse data; and the discussion section where you interpret your data and write up your findings.

METHODOLOGY

The objective of the methodology section is simple; it describes the research plan that you adopted when researching the topic so that the readers know and have confidence in the way the data was obtained and analysed. Therefore, it requires a clear and precise description of how the research was conducted. Description of the methodology section usually includes the following subheadings.

- Respondents
- Method of data collection
- Method of data analysis

Respondents

As a researcher, you need to describe the respondents that you have chosen so that the reader has a clear idea of who is being studied. Some of the characteristics you may need to describe include the number of participants that were used and their background, for instance:

- age range
- gender
- educational and professional qualifications
- work experience
- their area of specialisation

You also need to state how you chose your respondents and why. In short, you should provide any information you think is important about the characteristics of the respondents who helped you in the research.

Method of Data Collection

It is important to state your data gathering methods, for instance, whether you used primary data gathering methods (such as surveys, interviews, and observations) and/or obtained your data from secondary sources (like magazines, newspapers and the electronic sources). You need to explain why these methods were selected so that readers can be assured that the data collected is valid and trustworthy. It is also important to include all relevant information pertaining to the procedure/s

you followed when administering these data gathering methods, like the place where you administered the tool and even the time when the tool was administered.

If you are using a survey, interview or observation method, include in the Appendix a copy of the actual survey form, the interview questions or the observation sheet. This enables your audience to refer to them for a clear understanding of the tools used.

Method of Data Analysis

In this section, you need to describe the methods and tools used to analyse the data that have been collected, so that once again the audience is assured of the credibility of the findings. You also need to justify your choice of methods and tools.

ACTIVITY 8.1

Below is an extract from the *Method* section of a report. Read the text and answer the questions that follow.

> The participants were 60 undergraduates, 30 of whom were female students. They were selected according to a stratified random sampling method so that there was a proportionate representation of students from Year 1, Year 2, Year 3 and Year 4 from the different faculties in the university. 60 questionnaires were used to obtain the opinion of the undergraduates regarding the facilities provided in the main library (see Appendix A for a copy of the questionnaire). We also interviewed the head of the library unit and the librarians to obtain clarification about the facilities provided by the library. The data gathered via the questionnaires were analysed quantitatively using statistical tools (SPSS version 17) while the data obtained via the interviews were analysed qualitatively using NVivo.

1. How many respondents were used in the study?
2. Who were the respondents? Give details.
3. How were they selected?
4. Why were they selected in this manner?
5. What were the data gathering tools used in this research?
6. How was the data analysed?

Once the writing up of the method section is completed, the next stage in the drafting process is the interpretation of the data.

INTERPRETATION OF DATA

This section contains the main thrust of your argument and is usually known by any one of these headings: *Discussion, Findings, Results* or *Details*. This section is

the most challenging for many report writers because it requires them to interpret the abundant data that has been gathered. One of the best ways to begin is by looking at the outline that you have constructed as this will serve to guide you (see Chapter 5).

It is advisable to begin with an introductory preview which will guide the readers in their understanding of the contents of this section. The introductory preview explains how the writer plans to organise and present the data gathered. It includes the topics this section will discuss, the order in which these are presented, and in some cases, the logic for the order. The introductory preview aligns the purpose of the study to the interpretation of the data and this helps the reader to better understand the message in the report. Not only does it ensure that the reader is clear of what to expect in this section but it also helps the writer to focus.

Below is an example of an introductory preview taken from a recommendation report.

3.0 Discussion

A comparison of data on cost, safety, and dependability serves as the basis for the decision as to which car Berjaya Car Rentals should purchase. Each of these criteria are applied across three makes of vehicles: Toyota, Hyundai and Honda. *The cost section* compares initial and trade-in-values, operating costs and petrol consumption. *Next, the section on car safety* evaluates three safety features: anti-lock brake system, air bag and electronic stability control. *The third section* examines and compares repair records and time lost because of automobile breakdowns. *The section ends* with an evaluation and comparison of all these aspects in order to make the required recommendation.

It includes the topics that will be discussed: cost, safety and dependability.

It clearly states:
- *how these three main criteria are used to make the comparison;*
- *the breakdown of these criteria into their component parts; and*
- *the order in which the analysis is presented.*

(The introductory preview functions as an introduction to the discussion section of the report and, therefore, does not come under a separate heading.)

ORGANISING THE INFORMATION IN THE BODY OF THE REPORT

There are six aspects that need to be borne in mind when organising and writing up the body of the report. These comprise:

1. using appropriate headings and subheadings to organise information
2. presenting information using non-linear texts so that it is easily understood
3. presenting information in a coherent and cohesive manner
4. using plain, easy-to-understand language
5. ensuring that borrowed information is cited correctly in the text
6. making sure that information is cross-referred

This chapter will discuss the first and second aspects while two of the subsequent chapters will discuss the other aspects.

Headings and Subheadings

Headings and subheadings help readers by breaking up the report into manageable units. They assist in the following ways.

- Lead the reader through the report
- Make it easier for the readers to access the information
- Allow the reader, at a glance, to see the importance of the topics
- Help readers go straight to the required section(s)
- Enable the readers to see the logical order of the report
- Serve as visual breaks

It is important that the headings and subheadings are informative. They should have sufficient details to indicate to the reader(s) what the section is about.

For example, compare the two headings given below.

(a) Financial Knowledge

(b) Gauging the level of Personal Financial Knowledge

A heading like 'Financial Knowledge' is very general. The reader may interpret it in many ways, for instance, how to gain financial knowledge or how to measure financial knowledge.

In (b), making the heading informative by giving very precise information removes any kind of misinterpretations. Therefore, 'Gauging the level of Personal Financial Knowledge' allows the reader to predict the information clearly. The reader can predict that some kind of measurement or test is done to determine personal financial knowledge of the subjects in the report.

You also need to remember to arrange your headings/subheadings in an appropriate manner. Generally, the headings/subheadings are arranged in the following ways:

- Using chronological development, that is, from past conditions through to present conditions and projected future conditions
- Arranging them from most important to least important

In a problem-solution report, the headings are usually arranged from most important to least important.

When writing headings and subheadings, you need to ensure that they are parallel in structure throughout the report (see Chapter 5 for explanation about parallelism).

ACTIVITY 8.2

Read the report below and complete the blanks with appropriate headings/subheadings from the list given. Include appropriate classification numbers, for example 1.1.

• Land Misuse	• Restriction on Marriage Behaviour
• Remedying of Mistakes	• Conclusion
• Reducing the Rate of Loss of Non-Renewable Resources	• Release of Sterile Insects
• Control of Pest Organisms	• Perpetual Widowhood
• Biological Control By Means of Predators or Parasites	• Disposal of End-Products
• Maintaining Population Levels among Wild Animals	• Scope
• Contraception	• Fertility Control

CONSERVATION

1.0 Introduction

To *conserve* means to *maintain*, to *preserve*, to *keep the same*. *Conservative* means *resistant to change*, but in its modern usage, conservation means rather more than merely keeping things as they are. It is not simply a matter of preventing change. In any case, this is not possible. Its chief aim is to prevent further degradation of the environment, despite all the various demands being made upon it. Conservation is, in fact, good husbandry on both a national and world wide scale, with all the planning, foresight and cooperation that it entails.

The problem of conservation can be divided into three groups:

(a) Remedying of mistakes

(b) Reducing the rate of loss of non-renewable resources

(c) Controlling population

_____

In order for conservation to take place, one of the most important things to do is to resolve the mistakes that have been made. There are many conditions that need to be dealt with.

2.1 Wildlife

Where species have become entirely extinct, clearly nothing can be done. Where species may be threatened with extinction, a number of actions are possible, and it is an encouraging sign of changing attitudes that more and more is being done, for example:

(a) species are protected by law. The protection of Birds Act (1954) makes it an offence to take the eggs of, or interfere with, any wild bird.

(b) areas in which threatened species live are designated as nature reserves and subjected to rules whose sole aim is the preservation of species within them, for example, Brownsea Island in Poole harbour (a bird sanctuary) and Yellowstone park in America. A number of organisations in Britain exist for the sole purpose of conserving wild life, both plant and animal. Trees, for example, may not be felled without a good reason, and an overall responsibility for trees in Britain is that of the Forestry Commission which was set up in 1919.

2.2 Pollution

More progress has been made in combating atmospheric pollution than perhaps in any other aspect of conservation. This is scarcely surprising, since the quality of the atmosphere is something that affects us all immediately and directly. The great London smog of 1952, for example, was held to be responsible for the deaths of four thousand people.

Pollution originating from petrol and diesel rises as the number of vehicles on the roads increases. Legislation introduced in America in 1967 lays down standards for the amount of pollution allowable from exhausts of road vehicles, and there is little doubt that this will spread to other countries as their traffic densities rise, or as their car exports to America halt.

River pollution is also the target of concerted action. The Water Resources Act of 1963 set up a Water Resources Board acting through 29 river authorities, charged with the responsibility for the conservation and utilisation of water. Standards are laid down, enforceable by law, governing the permitted levels of pollutants which may be discharged into rivers. Similar regulations are applied in other countries – although accidents can occasionally happen, as was the case in 1969 when the illegal dumping of chemicals caused the destruction of millions of fish along the entire length of River Rhine.

_____

Land mismanagement at its worst calls up pictures of man-made deserts, the most ancient being the arid lands around the Mediterranean, and the most recent, the dustbowls of the American Mid-West.

As a result of legislation and actions by local authorities and private organisations, there appears to be a much greater sensitivity to the needs of conservation. Large stretches of the coastline are now protected from exploitation. The building of roads involves care to see that they take the route least likely to absorb good agricultural land. They are landscaped so as to disturb the appearance of the countryside as little as possible.

The first step in building a housing estate used to be the removal of trees. They are now realised to be an asset and are preserved. Planning, especially where central government is involved, is often seen as a restriction of liberty at the personal level. Such restriction is clearly necessary in the interests of all.

Most, if not all, modern technological societies which set the pace for the so-called emergent nations, are committed to the principle of a raised standard of living. This means a policy of increasing consumption of materials and a policy of continuous economic growth. In itself, this need not necessarily be harmful if it means that materials are re-cycled more rapidly. For example, it does not matter if a man changes his car for a new one every two years, or every six months, provided that its materials feed back into the cycle. It does matter, however, if the rate of accumulation of rusting cars increases four times as quickly. Part of the problem lies in the fact that the processing of raw materials into finished goods pays immediate dividends, whether it is a new suit or a new transistor radio. This incentive to possess creates the incentive to produce. There are no such incentives operating with goods that are worn out or discarded.

A serious problem is created by a modern society in the disposal of the end products of its industry, end products it is designed to create. Such society generates its own special kind of pollution. Modern advertising speaks glibly of the virtues of disposable packaging, meaning bottles, cans, plastic containers of various sorts and styrofoam containers. Disposable they may be, in the sense that they are virtually indestructible. An apple core tossed into the hedge from a passing motor car is rapidly decomposed, its materials being recycled. A plastic bottle, similarly disposed of, remains there for a long time.

Long term planning must therefore embody as part of its aims the establishment of a stable system, which continuously recycles its materials. It must:

(a) reprocess its end products. If it is possible to bring together a large number of complex materials to create, for example, a motor car, it is equally possible to create the means of reversing the process.

(b) develop other sources of power which are not exhaustible.

There are some signs that moves are being made in this direction. In some parts of the country, old cars can be collected for a nominal fee and removed to a depot, compacted and then dispatched to factories which recover metal parts. The latest types of garbage disposal plants are systems for recovering useful materials together with an end product which is germ free and odourless and can be used to improve the quality of the soil.

There seems little doubt that electricity will be the universal energy source of the future. There are no pollutants as a result of its use except in its production by oil or coal fire power stations. These are already being replaced by power stations run by nuclear reactors, although these also have waste disposal problems. The problems of pollution are already stimulating research into the development of lightweight, long-life electric batteries to replace the internal combustion engine as a portable source of power.

4.0 Population Control

This involves finding solutions to three quite separate problems:

(a) Control of those organisms (pests) whose numbers *increase* as a result of man's activities.

(b) Maintenance of numbers of organisms whose numbers *decrease* as a result of man's activities.

(c) Control of human population

Pest organisms increase in numbers because conditions are created which favour their way of life. Wholesale, massive spraying of crops is clearly an ineffective way of re-establishing a balance because (a) it has harmful effects on other organisms and (b) it does not, except in the short term, deal satisfactorily with the pest. Three different techniques are likely in the long term to replace pesticides.

Two spectacular examples of success in this field may be cited. The prickly pear cactus, accidentally introduced into Australia in the early nineteenth century, rapidly became a pest. It was eventually brought under control by the introduction of a moth, native to the Argentine. A serious pest of citrus fruit trees in California, the cottony cushion scale, also an immigrant from overseas, was ultimately controlled by the importation of its natural predator, a small ladybird.

4.1.2 Insect Trapping

A great deal of research is being carried out on studies of chemical substances produced by insects. Males are normally attracted to the females by their scent. Extractions or synthesis of compounds which produce the scent can serve as a basis for baiting the trap. Other promising lines of research involve identification and extraction of the compounds in the plant food which attract the insect. These also can be used as bait.

A large number of organisms are bred artificially in the laboratory, and subjected to radiation to render them sterile. They are then released in the field. When normal wild type insects mate with them, no fertile eggs are produced. The end result is a reduction in fertility of the pest organism.

As a nomadic hunter, for example, the American Indians, Man's role in a natural ecosystem is to act as a predator. He and His prey act as checks upon each other, each to some extent regulating the population of the other. With the development of a pastoral way of life and the domestication of animals, the prey, in effect, passes under complete control by Man. Both the rate of reproduction and the rate of killing is regulated. Many animal species still stand in the role of hunted prey in relation to Man, particularly marine animals. Wild species need, therefore, to be managed. For example, the herring catch needs to be regulated in terms not of the size of the market, but of the density of herring populations. The problems here require that international agreements be made, since the seas are not subject to national control.

4.3 Human Population Control

Although a great deal is known about population regulation in other species, there is very little positive understanding of the biological factors controlling fertility in human beings. Animal populations are controlled by external factors operating upon internal mechanisms, but the situation with human beings is complicated by the fact that we have a personal, or subjective view also to take into account. Human beings, too, appear to have a range of choice of actions denied to animals.

4.3.1 Age of Marriage

This is often some years after the attainment of sexual maturity.

4.3.2 Abortion and Infanticide

Many tribes practice regular abortion or infanticide as a method of population control.

_____

A widow in some societies is not allowed to remarry.

4.3.4 Restriction of Marriage Behaviour

Some tribes have laws which restrict the living together of men and women during certain seasons.

_____

Many societies have their own particular herbal mixtures or practices which are supposed to reduce fertility. Usually these are of doubtful value. If they work, it is by faith rather than function. Population changes in advanced modern industrial countries can be illustrated by reference to Britain where conditions are in many ways similar to those in other European countries. The pattern in North America is rather different because of large scale immigration, and in Ireland, where mass emigration has occurred.

_____

A modern community has a wide range of methods available to it for the purpose of deliberate fertility control, most of which have drawbacks of one sort or another.

_____

Many of the problems of conservation on the larger scale are problems which lie beyond the scope of action by the individual. Nevertheless, the sort of action a government may take in a democratic community is largely determined by what its people will accept, or, more positively, by what they want. The basis for constructive action must clearly be informed and educated public opinion. Even in these days, biology is still too often thought of as a study of buttercups, or the dissection of guinea-pigs, and a surprisingly large number of people have no idea how important biology is and how it affects their lives. It is still too often just another subject studied by young people who see no relevance in it other than as an examination subject. The possession of an ordinary level pass in biology is no guarantee that its owner will throw less litter about, or uproot fewer bluebells. It is an extraordinary fact that every year, millions of people visit the national parks, such as Dartmoor or Snowdonia, or the coast, in order to enjoy their beauty, yet see no inconsistency in leaving such places strewn with the litter of modern living: beer bottles, cigarette cartons, ice cream wrappers and so on. Learning biology means more than just learning facts. It also means applying them with sense and understanding. It isn't only the scientist who often seems to stand apart from the object of his studies. A large number of ordinary people also forget that they are a part of the environment in which they live. The world is our home and we should not disfigure or abuse it, for we shall be the losers in the end.

Source: Adapted from Brewer, C. V., & Burrow, C. D. (1972). Life: Form and Function – An Introduction to Biology (pp.270–276). London & Basingstoke: Macmillan Education Ltd.

Organising Information Using Non-Linear Texts

Information in a report must be presented in a manner that is easily understood by the audience and one way of doing this is by using non-linear texts. Non-linear texts are usually used:

- to organise quantitative information that cannot be clearly explained in words
- to emphasise important details concerning trends, characteristics, concepts and relationships among several variables
- to show procedures and processes visually
- to summarise information
- to provide visual breaks for the reader

The advantages of using non-linear texts are that they are:

- visually appealing
- easier to understand and remember
- more communicative, clearer and forceful than prose on its own

There are many kinds of non-linear texts that can be used to summarise and present information. Different types of non-linear texts can present the same idea, but only one would be most effective. For example, both bar graphs and line graphs can present trends, but the latter is more effective as it clearly shows the flow, or the movement. Thus, it is important to select the non-linear text that is most appropriate for your purpose.

The examples shown below are some of the non-linear texts commonly found in a report.

Tables

Tables are highly versatile and widely used. They are used to compare and categorise large amounts of information in an easy-to-read format.

Example

Table 8.1: Vehicles Involved in Road Accidents (January to March 2007 and 2008)

Consumer Category	2007 JAN-MARCH	2008 JAN-MARCH	DIFFERENCE	%
Car Driver/Passenger	287	308	21	7.3
Motorcycle Rider/Pillion	866	992	126	14.5
Pedestrian	157	187	30	19.1
Bicycle Rider/Pillion	47	45	-2	-4.3
Bus Driver/Passenger	16	6	-10	-62.5
Lorry Driver/Attendant	51	52	1	2.0
Van Driver/Attendant	27	17	-10	-37.0
4 Wheel Driver/Attendant	14	21	7	50.0
Other Vehicles	17	15	-2	-11.8
TOTAL	**1,482**	**1,643**	**161**	**10**

Source: Royal Malaysian Police Department

Bar graphs (single or multiple)

Bar graphs are used to make quick comparisons of data, to present differences and to depict trends.

Example

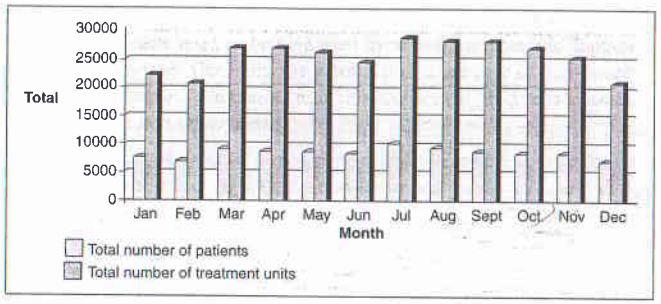


Figure 8.1: Total Number of Patients and Treatment Units in the Department of Physiotherapy (2009)

Pie charts

In pie charts, the entire circle represents a whole that amounts to 100 per cent. It shows how the whole is divided into parts and how the parts are related to the whole. Each sector is proportional in size to the amount each sector represents; therefore, it is easy to make generalisations and comparisons.

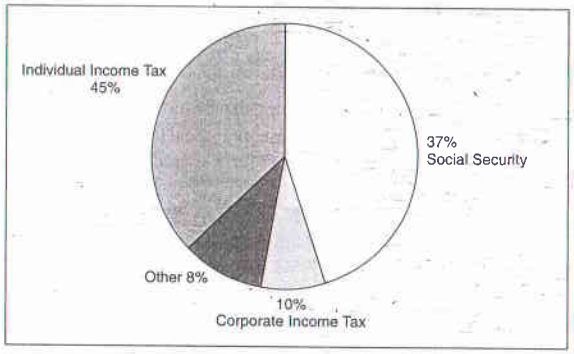


Figure 8.2: Revenue Sources for the Federal Government (20__)

Line graphs

Single or multiple line graphs are used to show and compare trends or cycles.

Example

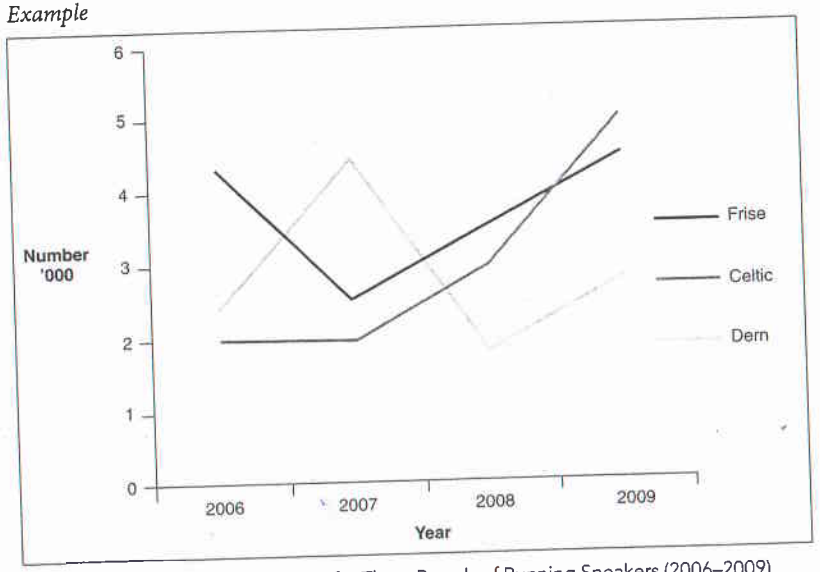


Figure 8.3: Production Figures for Three Brands of Running Sneakers (2006–2009)

Photographs

Photographs are used when you want to show something that cannot be depicted in any other non-linear form. The photograph below presents a visual impact of water pollution: the impact of this cannot be seen if any other form is used. Photographs or pictures are visual proof.



Figure 8.4: Destruction of the Natural Ecological System

There are several basic guidelines that a writer must follow when drawing a non-linear text.

1. Non-linear texts need to be identified by a label, number and caption (see Figure 8.5 below). Generally, the word *Figure* is used for all non-linear texts that are not tables. When using numbers, the normal practice is to use Arabic numbers for non-linear texts.
2. The title must be descriptive; you need to include precise information so that the reader knows exactly what the non-linear text refers to. You are advised to use wh-questions when writing the caption. The placement of the title is up to the writer: it can be placed above or below the non-linear text, but once you decide on the placement, make sure that it is consistent throughout the report.
3. Non-linear texts must be neat and simple. A neat graph is easy to read and interpret. Leave enough white space to make the non-linear text look uncluttered; also, it should be large enough.
4. Make sure that the information in the non-linear text is complete.

   For example:

   - when drawing a bar graph, the axes must also be labelled
   - provide a key/legend if it is necessary

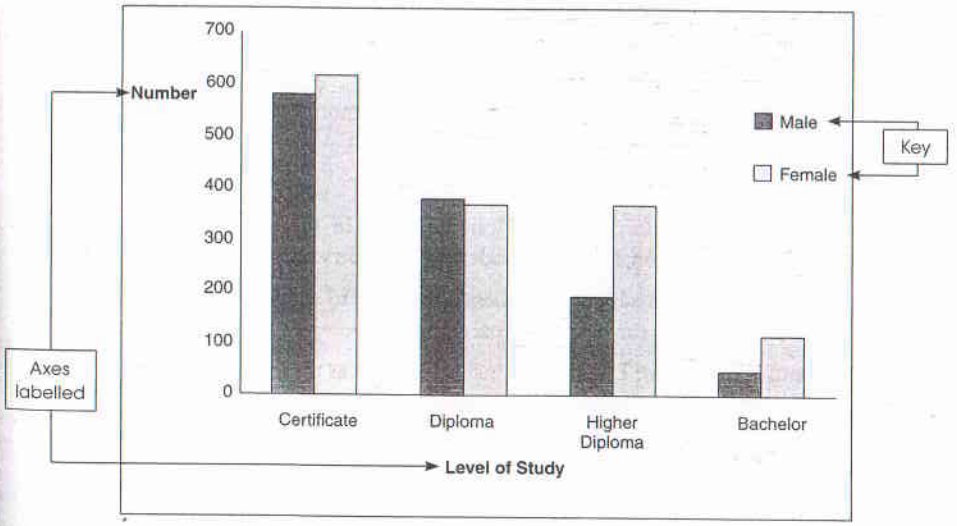


Figure 8.5: Comparison of Graduates by Level and Gender

Label Number Caption/Title

ACTIVITY 8.3

Study the table below that shows world crude oil and natural gas reserves in a particular year and do the tasks that follow.

Region	Crude Oil		Region	Crude Oil	
	Gas	Oil		Gas	Oil
	Billion Barrels			Billion Barrels	
North America	212.6	58.2	Middle East	739.15	722.6
South America	102.8	77.1	Africa	113.7	111.6
Europe	15.8	14.4	Asia	30.5	29.6
Eurasia	60.3	123.3	Oceania	1.7	4.4

(a) Draw a non-linear text that shows the proportion of the total oil and gas reserves based on regions.
(b) Write a suitable caption for the non-linear text that you have drawn.

Interpretation of non-linear texts

Non-linear texts used in the report must be explained for them to be effective. There are a number of points that need to be borne in mind when interpreting a non-linear text:

- Organise the information in terms of the common Introduction-Body-Conclusion (IBC) format.
- Keep to the scope of your task.
- Do not just repeat information that can be found in the non-linear text – interpret the information, and direct readers to what you want them to look at.
- Use figures/statistics to support statements. Evidence to support statements or additional information can be given in parentheses.
- Use rounded figures (for example, 2.0 instead of 2.33) as these will be easier for the readers to access and remember, unless giving specific, precise information is necessary.
- Use approximations if necessary to make ideas more accessible, for instance, 'more than 70 %'/'about 70%'.
- Do not draw inferences that cannot be supported by the text.
- Be consistent with the use of tenses which, incidentally, are usually in the past or present.

- Use hedging devices (*seems, appears to be, apparently, could, would, perhaps, etc.*) when necessary.
- Acquire the specialised vocabulary used for commentating on texts, such as, *significant, constant, stable, trend, proportion, gradual rise/fall, etc.*
- Vary the sentence structure to make the text interesting to read.

Given below is a table on students' results in the 'Sijil Tinggi Pelajaran Malaysia' examinations (2008-09). The description which follows illustrate the use of the guidelines given above to interpret and describe non-linear texts. For the purpose of exemplification, the focus is only on the mathematics and science results.

Table 8.2: 'Sijil Tinggi Pelajaran Malaysia' Results (2008–2009)

Subject	2008		2009		Percentage Difference
	No. of Candidates	Percentage Pass	No. of Candidates	Percentage Pass	
Tamil Language	730	53.30	595	58.10	+4.80
Malay Language	31,550	58.90	27,925	60.20	+1.30
Chinese Language	1,360	62.50	1,117	58.67	-3.83
Malay Literature	16,590	44.40	13,123	47.10	+2.70
Arabic Language	3,690	56.66	3,327	59.15	+2.49
English Language	144	65.45	125	60.30	-5.15
Economics	18,248	47.95	17,178	51.55	+3.60
History	31,687	53.87	26,755	54.54	+0.67
Usuluddin	4,145	62.15	3,740	59.96	-2.19
Syariah	5,721	60.40	5,040	61.20	+0.80
Accounting	2,355	58.80	2,235	60.80	+2.00
General Studies	40,900	73.80	38,765	74.90	+1.10
Physics	4,835	75.75	7,015	75.87	+0.02
Chemistry	4,886	84.10	7,065	82.90	-1.20
Biology	3,400	77.70	4,580	76.40	-1.30
Further Mathematics	1,420	75.60	2,270	74.00	-1.60
Subsidiary Mathematics	2,640	71.70	2,660	69.90	-1.80
Additional Mathematics	4,952	70.97	7,075	71.60	+0.63
Geography	12,670	62.20	13,155	61.80	-0.40
Art	5,658	74.70	4,695	77.67	+2.97

Table 8.2 refers to the results for 'Sijil Tinggi Pelajaran Malaysia' for the years 2008-2009. There is only a minimal (a range of only 2.43%) difference in the students' results in the Mathematics (Further, Subsidiary and Additional Mathematics) and Science (Physics, Chemistry and Biology) subjects between the years 2008 and 2009, although there is a dramatic increase in the number of students registered for all the subjects except Subsidiary Mathematics. The best performances are in Chemistry, with more than 82 percent passes for both years. Subsidiary and Additional Mathematics appear to be the worst. However, generally, more than 70 percent of the students have passed these subjects in both years. (Total Number of Words: 93)

Introduction – makes reference to the text and its contents

Body – identifies the main message in the text and supports it with evidence from text

Conclusion – end with a summarising comment that captures the main message

ACTIVITY 8.4

Below is a table which compares the cosmetic procedures undergone by men and women in a particular year. The interpretation and description is not well written. Improve it using the guidelines.

Table 8.3: Cosmetic Procedures Undergone by Men and Women

Procedures	Men	Women
Liposuction	357,980	398,848
Botox	329,519	445,656
Eyelid Surgery	32,564	208,199
Laser Hair Removal	185,684	226,974
Microdermabrasion	83,748	83,910

Table 8.3 shows the top five cosmetic procedures undergone by men and women in 2009. A total of 756,828 patients underwent liposuction. 52.7% of them were women while men accounted for 47.3%. 775,175 patients underwent Botox. Out of this number, 57.5% were women while 42.5% were men. Out of 240,763 patients who underwent eyelid surgery, 86.47% were women while 13.53% were men. A total of 412,658 patients underwent laser hair removal. 55% of them were women while 45% were men. 167,658 patients underwent microdermabrasion. Out of this number, 49.9% were male and the rest were women. Generally, more women went for cosmetic procedures compared to men in 2009. This shows that women care more for their looks than men.

WRITING SUPPORT 8.1

When describing trends in a graph, we can use verbs to talk about upward, downward or horizontal movements. Most verbs also have noun forms. They are generally the same.

Example

to rise (verb) – a rise (noun)

to fall (verb) – a fall (noun)

to fluctuate (verb) – a fluctuation (noun)

Sometimes we need to give more information about a trend, usually about the degree or movement of change.

Example

- The year started with a *sharp* decline in sales and *stabilised* in September.
- Sales increased *slowly* during January and then declined *steadily* until the end of the financial year.

Adverbs and adjectives can be used to modify verbs and nouns. Adverbs modify the verbs and usually end in 'ly' (to increase *significantly*), and adjectives modify nouns (a *significant* increase).

The following table shows some of the common words/expressions used when describing non-linear texts.

Table 8.4: The Language of Non-Linear Texts

Prepositions	*between* 1995 and 2000 ; *from* 1995 to 2000
Movement: Down	fell; declined; dropped; decreased; sank; went down; slid
Movement: Up	rose; went up; increased; grew; bounced back; rallied; recovered; soared
No Movement	remained steady; were unchanged; did not change; remained constant; remained stable; stabilised, flattened out; levelled off
Strong Words	shot up; rocketed; surged; plummeted; plunged; dived
Nouns	a flattening out; a stabilisation; a levelling off; a fall; a fluctuation
Adjectives	slight; a little; a lot; sharp; sudden; steep; dramatic; gradual; gentle; steady; substantial; considerable
Adverbs	Respectively; in that order; correspondingly

Tops and Bottoms	reached a peak; peaked; reached the highest; level fell to a low; sank to a trough; reached the bottom
Moving up & down	Fluctuated; moved unsteadily

The interpretation below demonstrates how the language of graphs can be used to describe trends.

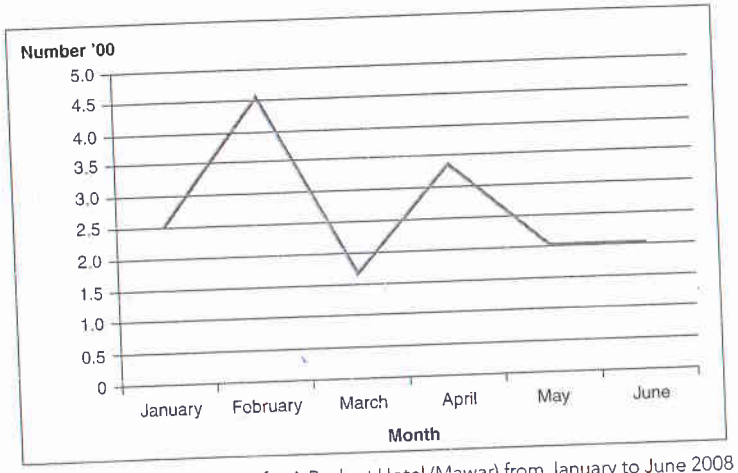


Figure 8.6: Occupancy Rate for A Budget Hotel (Mawar) from January to June 2008

Figure 8.6 shows the fluctuation in the occupancy rate of Mawar, a budget hotel, for the months of January to June. The busiest period is the first part of the year. There is a sharp increase between January and February, with 450 people staying at the hotel. After this the numbers drop sharply to less than 200 from February to March. Between March and April the numbers rise steeply, but decrease to 200 and remain at a plateau from May to June. Overall, the graph shows that the occupancy rate is highest at the beginning of the year.

Exercise 1

Study the graph below and compare, in less than 80 words, the sales volume for the three major exporters of batteries.

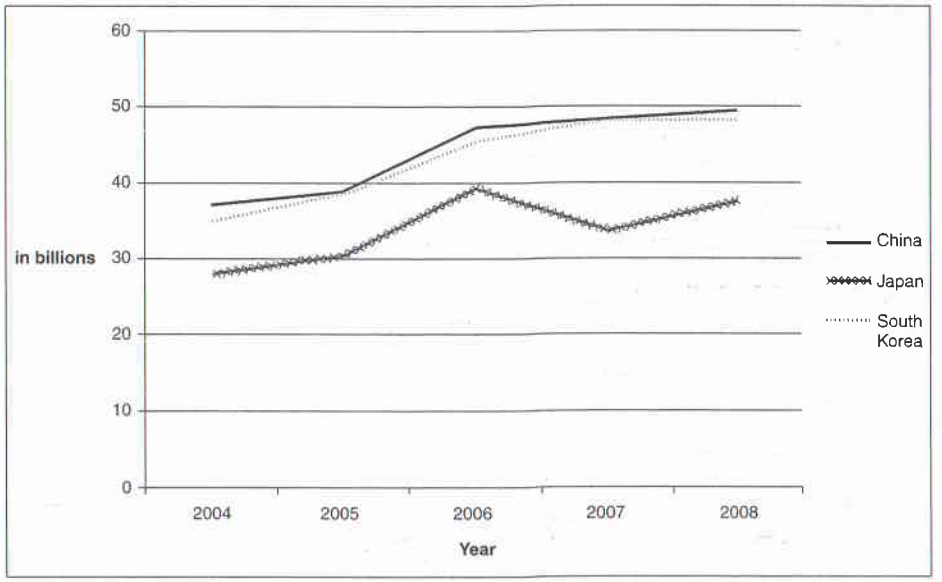


Figure 8.7: Sales Volume of Batteries for Three Major Exporters, 2004-2008

Exercise 2

Study the two tables below, and fill in blanks 1 to 10 in the text that follows based on the information in the tables.

Table 8.5: Social Problems Affecting Young People between the Ages of 13 and 24 (2004)

Without Parents' Knowledge	Boys (%)	Girls (%)	Total (%)
Going Out Until Late at Night	22.2	4.6	13.5
Gangsterism	1.9	0.9	1.4
Illegal Racing	4.8	1.0	2.9
Smoking	18.5	0.9	9.8
Drug Addiction	1.4	1.2	1.3

Table 8.6: Going Out Until Late at Night and Smoking

Without Parents' Knowledge	Going Out Until Late at Night		Smoking	
	1994	2004	1994	2004
Boys	3.7	22.2	6.2	18.5
Girls	0.8	4.6	0.2	0.9
Cities	2.1	13.0	2.7	8.7
Rural Areas	2.5	14.4	3.7	11.5

Source: The New Straits Times, July 12, 2006

Tables 8.5 and 8.6 show the statistics for some of the problems that are prevalent among youngsters between the ages of 13 and 24. These are the results of a survey carried out once every (1) _____ years since 1974 by the *National Population and Family Planning Board, Malaysia*. The next survey in 2014 will hopefully present more encouraging statistics.

According to the survey, two of the (2) _____ serious problems among young people are ***Going Out Until Late at Night*** and ***Smoking***. Both of these together account for (3) _____ % of the problems among the males. Among the females, they account for slightly (4) _____ than one-seventh of this percentage. Another more serious problem among the (5) _____ is ***Drug Addiction***, with (6) _____ following close behind. ***Gangsterism*** ties with ***Smoking*** at 0.9%. ***Drug Addiction*** appears to be the least serious of the problems among the males, as well as among (7) _____ taken as a whole.

According to (8) _____, it appears that youngsters are becoming increasingly rebellious. Among the four categories (***Boys, Girls, Cities and Rural Areas***), the greatest difference can be seen in the category of (9) _____, with ***Rural Areas*** occupying second position. Surprisingly, ***Rural Areas*** seems to have a (10) _____ percentage of problems with regard to ***Going Out Until Late at Night*** and ***Smoking*** compared to ***Cities***.

The Board, however, is optimistic that most of the problems faced by the youngsters can be overcome if their sources could be determined.

WRITING SUPPORT 8.2

Captions and headings are normally written as noun phrases, and not as complete sentences.

A **noun phrase** is either a single noun (or pronoun) or a group of words containing a noun or a pronoun. A noun phrase has a noun as its Head (main noun/subject). Sometimes adjective phrases or noun modifiers (that describe the head noun) come before the head noun. Prepositional phrases too often qualify the head noun. (Prepositional phrases will begin with a preposition and end with a noun, pronoun, gerund, or clause, the "object" of the preposition). Sometimes a caption can contain a number of prepositional phrases. It is also interesting to note that in captions and headings, the head noun is often a gerund (see Chapter 4 for a discussion of gerunds).

Examples

Adj. *H.Noun* *Prep. Phrase*

(a) Sexual Harassment in the Civil Service

H. Noun *Prep. Phrase* *Prep. Phrase* *Prep. Phrase*

(b) Comparison of Income between Men and Women in the Manufacturing Industry

H. Noun *Prep. Phrase* *Prep. Phrase* *Prep. Phrase*

(c) Intake of Students into Public and Private Universities in Malaysia (2009)

The examples above can function as either headings/subheadings or captions of non-linear texts.

Exercise 3

Study the sentences below, and convert them into possible captions/headings. The first has been done for you as an example.

(a) This is the process for getting a permit.
Getting a Permit

(b) The table shows how a renewal should be arranged.

(c) The imbalance in the study is due to weight gain.

(d) The figure shows the production value of transgenic orchids in Malaysia for the years 2004 to 2007.

(e) There is a need to improve the wireless service at *Coffee Station* to attract customers.

(f) The new incentives are introduced to protect the automotive industry in Malaysia.

(g) There is an increasing trend to invest in large scale bio-mass plants in Asia in 2007.

(h) There are many ways to manage stress in the workplace.

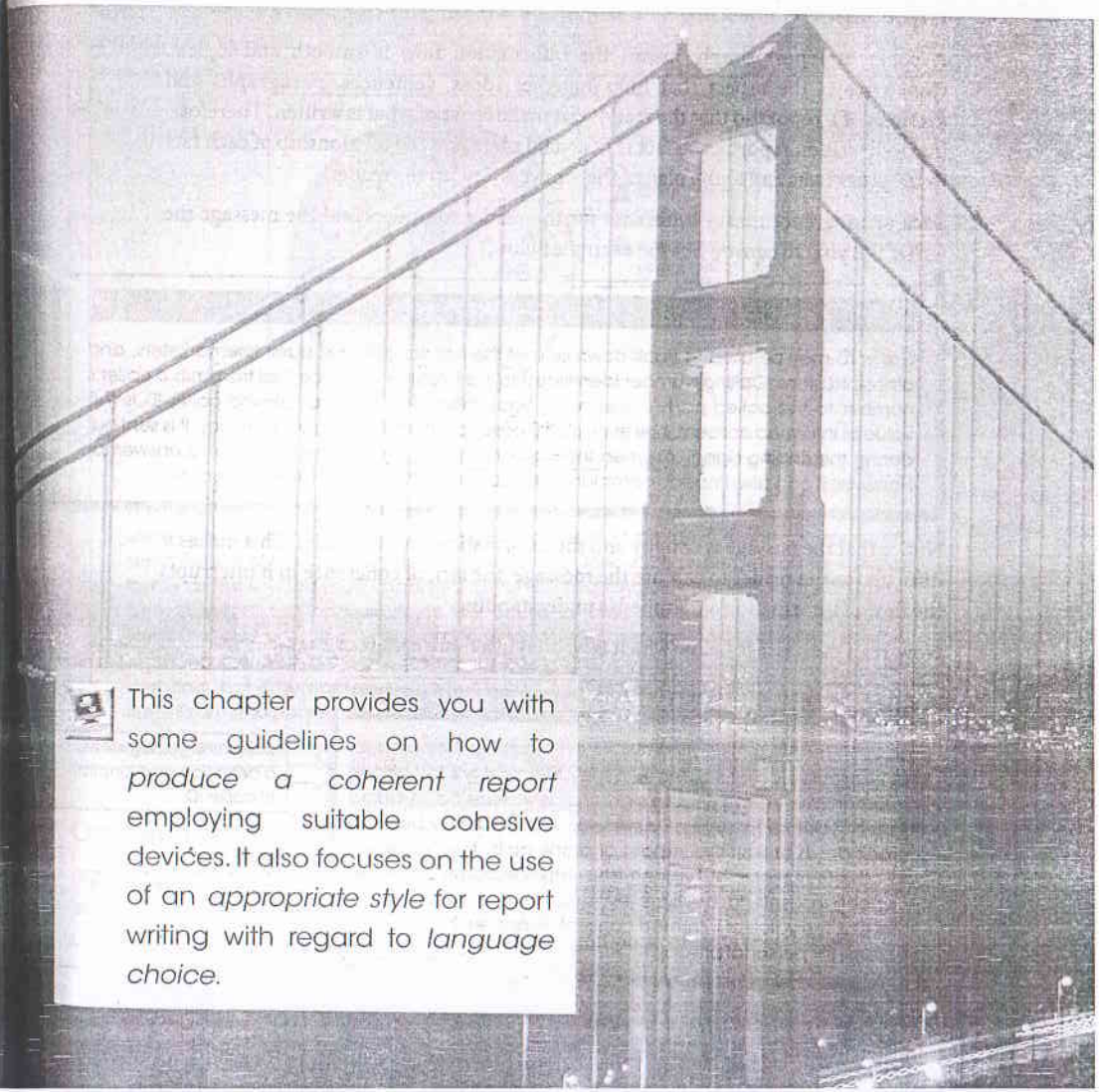
(i) The table shows the statistics in relation to access to education and class size, 1990–2005

(j) The figures show the world economic outlook from 2001–2010.

Sarguman, R et. al (2010) Report writing for Business & Professional Purposes. KL: McGraw Hill

CHAPTER 9

Drafting the Report (III): Coherence & Cohesion



This chapter provides you with some guidelines on how to *produce a coherent report* employing suitable cohesive devices. It also focuses on the use of an *appropriate style* for report writing with regard to *language choice*.